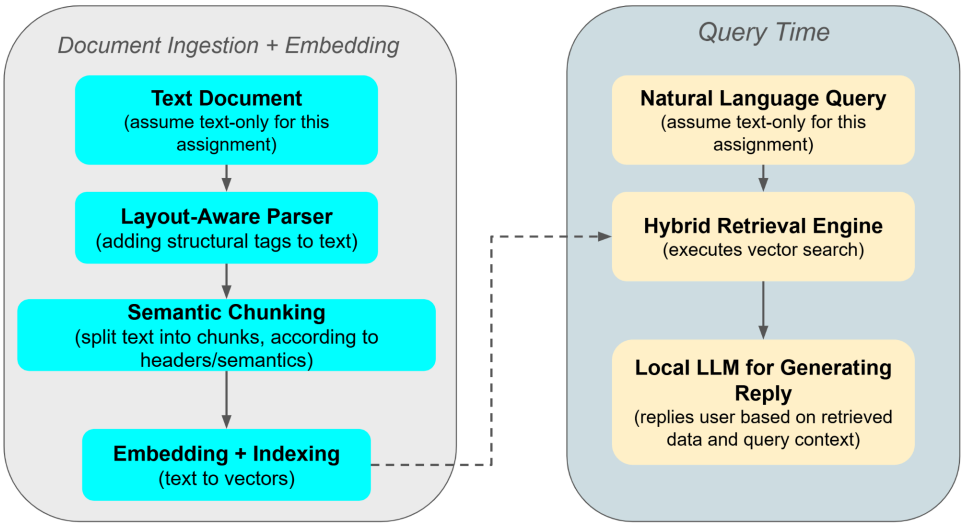
Deployment Team Take-Home Assignment Part A

Ng Zheng Xun

Assumptions:

- Internal Documents being uploaded are converted to Text Data
- Queries by Users are in Text only
- Documents being uploaded are sensitive in nature, should not be uploaded to external cloud providers/outside of company

Architecture Overview:



Key Decisions + Tradeoffs:

For the document ingestion side of things, the general RAG pipeline applies (chunking → embedding). However, a few additional optimisations are applied:

- Additional Layout-Aware Parser before Chunking (eg Docling or Unstructured), which intelligently tags text as paragraph, section heading, equation etc, for better retrieval / embedding
- Semantic Chunking of Parsed Text, which chunks text not based on an arbitrary character limit but through capturing an entire semantic point or document section
- Trade off: the above would result in longer embedding times for the 2000 documents uploaded per month. However, it is a one-time process per document.

On the query side of things, we make a choice to:

- Finetune an open-source LLM that can be hosted on-prem, due to the sensitive nature of documents. Trade off: lower reasoning capabilities from queries, but much faster time-to-first -token and lower latency.

Post-Deployment Monitoring:

- **Answer Quality,** through evaluation metrics such as LLM-as-a-judge
- As latency is important to users, will monitor **Time-to-First-Token** and **Total Response Duration** (avg across several queries)

Failure Mode Surfaced During Production:

- While Semantic Chunking is implemented, some specific cases may still fall through the cracks, especially for highly specialised and company-specific terms and phrases. Therefore, a possible failure mode will be **Incorrect/Insufficient Chunking of the documents**.